

Calculus III — Exercise Sheet

Chapter 1: Differentiation | Chapter 2: Multiple Integrals

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1. Vectors and Geometry of Space

Exercise 1.1. Find the vector $\overrightarrow{AB}$ and its magnitude for each of the following pairs of points.

(a) $A(1, -2, 4), B(3, 0, -1)$.

(b) $A(-3, 5, 0), B(2, -1, 6)$.

Exercise 1.2. Let $\mathbf{u} = \langle 2, -1, 3 \rangle$ and $\mathbf{v} = \langle -1, 4, 2 \rangle$.

(a) Compute the dot product $\mathbf{u} \cdot \mathbf{v}$.

(b) Find the angle θ between $\mathbf{u}$ and $\mathbf{v}$.

(c) Compute the cross product $\mathbf{u} \times \mathbf{v}$ and verify that it is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$.

(d) Find a unit vector in the direction of $\mathbf{u}$.

Exercise 1.3. (a) Find parametric and symmetric equations of the line passing through $P(-1, 2, 3)$ and parallel to the vector $\mathbf{v} = \langle 3, -1, 5 \rangle$.

(b) Find parametric equations of the line passing through the points $A(1, 0, -2)$ and $B(4, 3, 1)$.

(c) Find the point where the line $x = 1 + 2t, y = -t, z = 3 + t$ meets the plane $2x - y + z = 7$.

Exercise 1.4. (a) Find an equation of the plane passing through $P(2, -1, 3)$ with normal vector $\mathbf{n} = \langle 1, 5, -2 \rangle$.

(b) Find an equation of the plane through the three points $P(0, 1, 1), Q(1, 0, 1)$, and $R(1, 1, 0)$.

(c) Determine whether the planes $2x - 3y + z = 4$ and $4x - 6y + 2z = 1$ are parallel, identical, or neither.

Exercise 1.5. Describe and sketch the following surfaces in $\mathbb{R}^3$.

(a) $x^2 + z^2 = 9$.

(b) $z = 4 - x^2 - y^2$.

(c) $\frac{x^2}{4} + \frac{y^2}{9} + \frac{z^2}{16} = 1$.

2. Functions of Several Variables

Exercise 2.1. For each function, (i) find and sketch the domain, (ii) determine the range, and (iii) describe three representative level curves.

(a) $f(x, y) = \sqrt{16 - x^2 - y^2}$.

(b) $f(x, y) = \ln(x + y - 1)$.

(c) $f(x, y) = \frac{x^2 - y^2}{x^2 + y^2}.$

Exercise 2.2. Find the domain and the level surfaces of

$$f(x, y, z) = \sqrt{9 - x^2 - y^2 - z^2}.$$

Exercise 2.3. Sketch the level curves of $f(x, y) = x^2 + 4y^2$ for the values $k = 0, 1, 4, 9$.

3. Limits and Continuity

Exercise 3.1. Evaluate each limit, or show that it does not exist.

(a) $\lim_{(x,y) \rightarrow (1,2)} (x^3y - 2xy^2 + 5).$

(b) $\lim_{(x,y) \rightarrow (0,0)} \frac{5x^2y}{x^2 + y^2}.$

(c) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - xy}{\sqrt{x} - \sqrt{y}}$ (where defined).

(d) $\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + y^4}.$

Exercise 3.2. Use the Squeeze Theorem to prove that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2y^2}{x^2 + y^2} = 0.$$

Exercise 3.3. Discuss the continuity of the function

$$f(x, y) = \begin{cases} \frac{2xy}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

4. Partial Derivatives and Higher Derivatives

Exercise 4.1. Find all first-order partial derivatives of each function.

(a) $f(x, y) = x^4y - 3x^2y^3 + 7y.$

(b) $g(x, y) = e^x \sin(xy).$

(c) $h(x, y, z) = xy^2z^3 + \ln(x + 2y + 3z).$

(d) $w(s, t) = s \arctan(t/s).$

Exercise 4.2. Find all second-order partial derivatives of $f(x, y) = x^3y^2 - 2x^2y^4 + 5xy$, and verify Clairaut's theorem ($f_{xy} = f_{yx}$).

Exercise 4.3. A function $f(x, y)$ satisfies Laplace's equation if $f_{xx} + f_{yy} = 0$. Verify that each of the following functions is harmonic.

(a) $f(x, y) = x^3 - 3xy^2.$

(b) $f(x, y) = e^x \cos y.$

(c) $f(x, y) = \ln(x^2 + y^2).$

Exercise 4.4. Let $f(x, y) = x^2e^y + y \sin x$. Calculate f_{xxy} and f_{xyy} .

5. Tangent Planes and Linear Approximations

Exercise 5.1. (a) Find the equation of the tangent plane to the surface $z = x^2 + 2y^2 - 3x$ at the point $(1, 1, 0)$.

(b) Find the equation of the tangent plane to the surface $z = e^{x-y}$ at the point $(2, 2, 1)$.

Exercise 5.2. Find the linearisation $L(x, y)$ of $f(x, y) = \sqrt{20 - x^2 - 7y^2}$ at $(2, 1)$, and use it to estimate $f(1.95, 1.05)$.

Exercise 5.3. The dimensions of a closed rectangular box are measured to be 80 cm, 60 cm, and 50 cm, with a possible error of ± 0.2 cm in each measurement. Use differentials to estimate the maximum error in the calculated volume of the box.

Exercise 5.4. Find the total differential dz for each function.

(a) $z = x^4y + x\sqrt{y}$.

(b) $z = \arctan\left(\frac{x}{y}\right)$.

6. The Chain Rule

Exercise 6.1. Use the Chain Rule to find dz/dt .

(a) $z = x^2y + 3xy^4$, $x = \sin t$, $y = e^t$.

(b) $z = \sqrt{x^2 + y^2}$, $x = e^t \cos t$, $y = e^t \sin t$.

Exercise 6.2. Use the Chain Rule to find $\partial z/\partial s$ and $\partial z/\partial t$.

(a) $z = e^{2x+3y}$, $x = s/t$, $y = t/s$.

(b) $z = x^2 \tan y$, $x = st^2$, $y = s^2t$.

Exercise 6.3 (Implicit Differentiation). (a) Find dy/dx if $x^4 + x^2y^2 + y^4 = 3$.

(b) Find $\partial z/\partial x$ and $\partial z/\partial y$ if $x^2 + 2y^2 + 3z^2 + xyz = 6$.

7. Directional Derivatives and the Gradient

Exercise 7.1. Find the directional derivative of each function at the given point in the given direction.

(a) $f(x, y) = x^2y^3 - 2x$, at $(1, -1)$, direction angle $\theta = \pi/4$.

(b) $f(x, y, z) = x^2yz + xz^2$, at $(1, 1, 1)$, $\mathbf{v} = \langle 1, 2, -1 \rangle$.

Exercise 7.2. (a) Find the gradient of $f(x, y) = \sin(x + 2y)$ and evaluate it at $(\pi/6, 0)$.

(b) Find the maximum rate of change of $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ at $(1, 2, 2)$ and the direction in which it occurs.

Exercise 7.3. The temperature at a point (x, y) in the plane is given by $T(x, y) = 200e^{-(x^2+y^2)/100}$. A bug is located at $(3, 4)$.

- (a) In which direction should the bug move to warm up most quickly?
- (b) How fast does the temperature change in that direction?
- (c) If the bug moves in the direction $\mathbf{u} = \langle 4, 3 \rangle / 5$, what is the rate of change of temperature?

8. Extrema of Functions of Several Variables

Exercise 8.1. Find and classify all critical points (local max, local min, or saddle point) of each function.

- (a) $f(x, y) = x^2 + xy + y^2 - 3x$.
- (b) $f(x, y) = 4xy - x^4 - y^4$.
- (c) $f(x, y) = e^x(x + y^2 + 2y)$.

Exercise 8.2. Find the absolute maximum and minimum values of $f(x, y) = x^2 - 2xy + 2y$ on the closed triangular region D with vertices $(0, 0)$, $(3, 0)$, and $(3, 3)$.

Exercise 8.3. Find three positive numbers a , b , c whose sum is 100 and whose product is as large as possible.

9. Lagrange Multipliers

Exercise 9.1. Use Lagrange multipliers to find the extreme values of each function subject to the given constraint.

- (a) Maximise $f(x, y) = xy$ subject to $x + 2y = 8$.
- (b) Minimise $f(x, y, z) = x^2 + y^2 + z^2$ subject to $x + y + z = 12$.
- (c) Find the extreme values of $f(x, y) = x^2 + 2y^2$ on the circle $x^2 + y^2 = 1$.

Exercise 9.2. Use Lagrange multipliers to find the point on the plane $2x + y + z = 5$ closest to the origin.

Exercise 9.3. A rectangular box without a lid is to be made from 12 m^2 of cardboard. Use Lagrange multipliers to find the dimensions that maximise the volume of the box.

10. Double Integrals over Rectangles

Exercise 10.1. Evaluate the following iterated integrals.

- (a) $\int_0^3 \int_1^2 (x^2y - 2xy^3) \, dy \, dx$.
- (b) $\int_0^\pi \int_0^1 y \sin(xy) \, dy \, dx$.
- (c) $\int_0^1 \int_0^1 xe^{xy} \, dy \, dx$.

Exercise 10.2. Evaluate $\iint_R (x+2y) \, dA$ where $R = [0, 2] \times [0, 3]$ in two ways: integrating first with respect to y then x , and vice versa. Verify you get the same answer.

Exercise 10.3. Find the volume of the solid bounded above by the plane $z = 2 + x + 3y$ and below by the rectangle $R = [-1, 1] \times [0, 2]$ in the xy -plane.

11. Double Integrals over General Regions

Exercise 11.1. Evaluate each double integral over the described region D .

- (a) $\iint_D (2x - y) \, dA$, where D is bounded by the curves $y = x^2$ and $y = x + 2$.
- (b) $\iint_D xy \, dA$, where D is bounded by $y = x$ and $y = x^2$.
- (c) $\iint_D x \, dA$, where D is the triangle with vertices $(0, 0)$, $(1, 0)$, and $(0, 1)$.

Exercise 11.2. Reverse the order of integration and evaluate.

- (a) $\int_0^1 \int_x^1 \sin(y^2) \, dy \, dx$.
- (b) $\int_0^4 \int_{\sqrt{y}}^2 e^{x^3} \, dx \, dy$.

Exercise 11.3. Use a double integral to find the area enclosed by the curves $y = 3x - x^2$ and $y = x$.

12. Double Integrals in Polar Coordinates

Exercise 12.1. Convert to polar coordinates and evaluate.

- (a) $\iint_D \sqrt{x^2 + y^2} \, dA$, $D = \{(x, y) : x^2 + y^2 \leq 4, x \geq 0\}$.
- (b) $\iint_D e^{-(x^2+y^2)} \, dA$, D is the disk of radius a centred at the origin.
- (c) $\int_0^2 \int_0^{\sqrt{4-x^2}} (x^2 + y^2)^{3/2} \, dy \, dx$.

Exercise 12.2. Find the volume of the solid that lies under the paraboloid $z = 9 - x^2 - y^2$, above the xy -plane, and inside the cylinder $x^2 + y^2 = 4$.

Exercise 12.3. Use polar coordinates to find the area of the region inside the cardioid $r = 1 + \cos \theta$ and outside the circle $r = 1$.

13. Triple Integrals

Exercise 13.1. Evaluate the following triple integrals.

- (a) $\int_0^2 \int_0^x \int_0^{x+y} 6xy \, dz \, dy \, dx.$
- (b) $\iiint_B x e^{y+2z} \, dV$, where $B = [0, 1] \times [0, 1] \times [0, 1].$
- (c) $\iiint_E y \, dV$, where E is the solid bounded by $z = 0$, $z = x + y + 5$, $x = 0$, $x = 1$, $y = 0$, and $y = 1.$

Exercise 13.2. (a) Evaluate $\iiint_E z \, dV$, where E is the solid tetrahedron with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1).$

- (b) Find the volume of the solid enclosed by the paraboloids $z = x^2 + y^2$ and $z = 36 - 3x^2 - 3y^2.$

14. Triple Integrals in Cylindrical Coordinates

Exercise 14.1. Use cylindrical coordinates to evaluate the following integrals.

- (a) $\iiint_E (x^2 + y^2) \, dV$, where E is the solid inside the cylinder $x^2 + y^2 = 4$ and between the planes $z = -3$ and $z = 5.$
- (b) $\iiint_E \sqrt{x^2 + y^2} \, dV$, where E is bounded above by $z = 4$ and below by the paraboloid $z = x^2 + y^2.$
- (c) $\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2}}^2 (x^2 + y^2) \, dz \, dy \, dx.$

Exercise 14.2. A solid hemisphere of radius a has its flat face in the xy -plane. Using cylindrical coordinates, find the volume of the hemisphere.

15. Triple Integrals in Spherical Coordinates

Exercise 15.1. Use spherical coordinates to evaluate the following integrals.

- (a) $\iiint_B e^{(x^2+y^2+z^2)^{3/2}} \, dV$, where B is the unit ball $x^2 + y^2 + z^2 \leq 1.$
- (b) $\iiint_H (x^2 + y^2 + z^2)^2 \, dV$, where H is the upper hemisphere of radius 2.
- (c) $\iiint_E z \, dV$, where E lies above the cone $z = \sqrt{x^2 + y^2}$ and inside the sphere $x^2 + y^2 + z^2 = 4.$

Exercise 15.2. (a) Find the volume of the solid that lies inside the sphere $x^2 + y^2 + z^2 = 4$ and above the cone $z = \sqrt{x^2 + y^2}.$

- (b) Use spherical coordinates to evaluate $\iiint_E (x^2 + y^2) \, dV$, where E is the ball of radius 3.

16. Mixed Challenge Problems

Exercise 16.1. The temperature in a metal plate is given by $T(x, y) = x^2 - xy + y^2$.

- (a) Find all critical points of T and classify them.
- (b) Find the maximum and minimum temperatures on the disk $x^2 + y^2 \leq 4$.

Exercise 16.2. Let $f(x, y) = x^2 + 4xy + y^3$.

- (a) Use implicit differentiation to find dy/dx if $f(x, y) = 0$.
- (b) Find the gradient ∇f at $(1, -1)$.
- (c) Compute the directional derivative at $(1, -1)$ in the direction toward the origin.

Exercise 16.3. (*Gaussian Integral via double integrals*) Using polar coordinates, evaluate $I = \int_{-\infty}^{\infty} e^{-x^2} dx$ by computing

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2} dx \right) \left(\int_{-\infty}^{\infty} e^{-y^2} dy \right) = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)} dA.$$

Exercise 16.4. Find the volume of the region E bounded by $z = x^2 + y^2$ (paraboloid) and $x^2 + y^2 + z^2 = 6$ (sphere) using the most convenient coordinate system, justifying your choice.

Exercise 16.5. A curve in the xy -plane is defined implicitly by $F(x, y) = x \sin(y) - y \cos(x) = 0$.

- (a) Use the implicit function theorem to find dy/dx near $(0, 0)$.
- (b) Write the equation of the tangent line to the curve at $(0, 0)$.

Good luck! Remember: mathematics is not a spectator sport.